

B EVALUATION DETAILS

B.1 HUMAN PERFORMANCE ASSESSMENT DETAILS

To establish a robust human baseline, we recruited 54 participants through a custom online platform. The process yielded 1,684 valid responses, with each of the 559 benchmark items being answered by an average of 3 participants.

The median response time across all human responses was 26.9 seconds, with a mean of 40.3 seconds. The difference suggests that a subset of questions required substantially longer deliberation, skewing the mean. A detailed breakdown of performance by task category, presented in Table 9, reveals a clear inverse relationship between response time and accuracy. The analysis highlights that the two

Table 9: Human Performance by Task: Accuracy and Response Time.

Task	DISE Category	Accuracy (%)	Mean Time (s)	Median Time (s)
3D Combination	E-D	56.4	59.2	34.8
2D Shape Finding	I-S	61.5	58.5	46.4
2D Combination	E-D	75.2	36.8	32.0
2D Folding	I-D	76.5	25.6	17.0
Fold and Punch	I-D	76.8	55.4	44.4
2D Rotation	I-D	78.1	40.4	31.5
3D Projection	E-S	81.1	28.0	20.8
3D Shape Finding	I-S	81.8	31.4	23.3
3D Rotation	I-D	82.0	29.8	21.8
3D Folding	I-D	86.6	44.4	33.6

tasks with the lowest human accuracy, 3D Combination (56.4%) and 2D Shape Finding (61.5%), are also the tasks that commanded the longest mean response times (59.2s and 58.5s, respectively). This empirically confirms that these tasks impose the highest cognitive load. The mental simulation required to assemble complex parts in 3D Combination (Extrinsic-Dynamic) and the demanding visual search needed to disentangle embedded figures in 2D Shape Finding (Intrinsic-Static) are inherently time-consuming and error-prone for humans, providing a quantitative justification for their difficulty. Conversely, tasks with high accuracy, such as 3D Folding and 3D Rotation, generally required less time, indicating a lower cognitive barrier.

In order to obtain an unbiased estimate of human baseline performance over the full item pool, we employ a matrix-sampling design in which each participant completes only a single booklet of K items out of the total pool of I items. Adjacent booklets share a small set of a anchor items (≈ 10). We recruited 54 participants for the study. Before participation, all individuals provided their informed consent and all procedures were conducted in accordance with relevant ethical guidelines. The data collection process yielded a total of 1679 valid responses across all items. Each item was answered by an average of 3 participants. The main paper reports human performance with Classical Test Theory (CTT) results, while Item Response Theory (IRT) is used for cross-validation.

Analysis Methodology The collected response data were analyzed using two psychometric frameworks:

CLASSICAL TEST THEORY (CTT) For each item booklet and for the anchor-linked “overall” pool, the proportion-correct statistic was computed as

$$\hat{p} = \frac{x}{N} \quad (1)$$

where x is the number of correct responses and N is the total number of responses to that booklet or pool. Sampling variability was quantified with the Wald standard error

$$SE_{CTT} = \sqrt{\frac{\hat{p}(1 - \hat{p})}{N}}, \quad (2)$$

yielding a two-sided 95% confidence interval (CI)

$$\hat{p} \pm 1.96 \text{ SE}_{\text{CTT}}. \quad (3)$$

ITEM RESPONSE THEORY (IRT) To cross-validate the CTT findings and place all items on a common latent-ability scale, we fitted a two-parameter logistic (2PL) model to the entire response matrix,

$$P_{ij} = \sigma[a_i(\theta_j - b_i)] = \frac{1}{1 + \exp[-a_i(\theta_j - b_i)]}, \quad (4)$$

where P_{ij} is the probability that participant j (ability θ_j) answers item i (discrimination a_i , difficulty b_i) correctly.

For a designated item subset (e.g., a DISE category) containing I items, the model yields an item-level expected probability of success $\bar{P}_i$. The category-level expected accuracy is then

$$\hat{p}_{\text{IRT}} = \frac{1}{I} \sum_i \bar{P}_i = \bar{P}. \quad (5)$$

Between-item variability was captured via the sample variance

$$s^2 = \frac{1}{I-1} \sum_{i=1}^I (\bar{P}_i - \bar{P})^2, \quad (6)$$

leading to the standard error

$$\text{SE}_{\text{IRT}} = \frac{s}{\sqrt{I}}, \quad (7)$$

and the 95% CI

$$\hat{p}_{\text{IRT}} \pm 1.96 \text{ SE}_{\text{IRT}}. \quad (8)$$

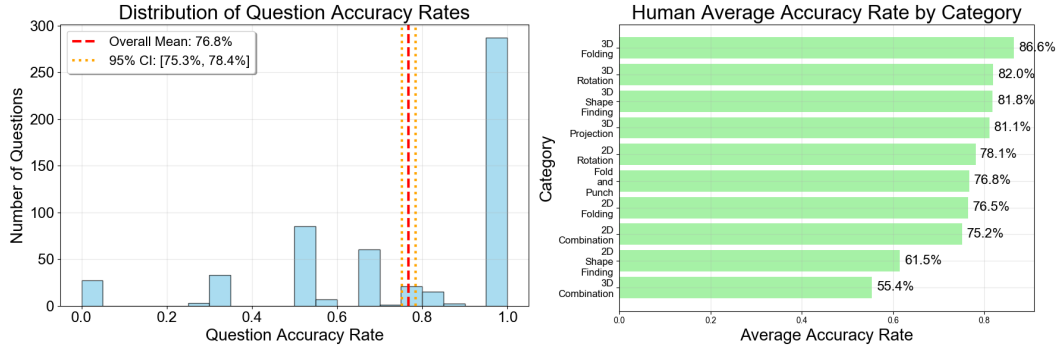
Results To provide a comprehensive view, we compare the results from both CTT and IRT analyses. Figure 10, Table 11 juxtaposes the observed accuracy from CTT with the model-based predictions and item parameters from IRT for each DISE category. This comparison highlights the synergy between the two methodologies. The CTT accuracy provides a direct, empirical measure of performance, while the IRT parameters offer an explanation for these results.

Table 10: Parameters of the Human Assessment

Parameters	Num.
Number of Participants	54
Total Number of Items I	559
Total Responses N	1679
Number of Booklets	19

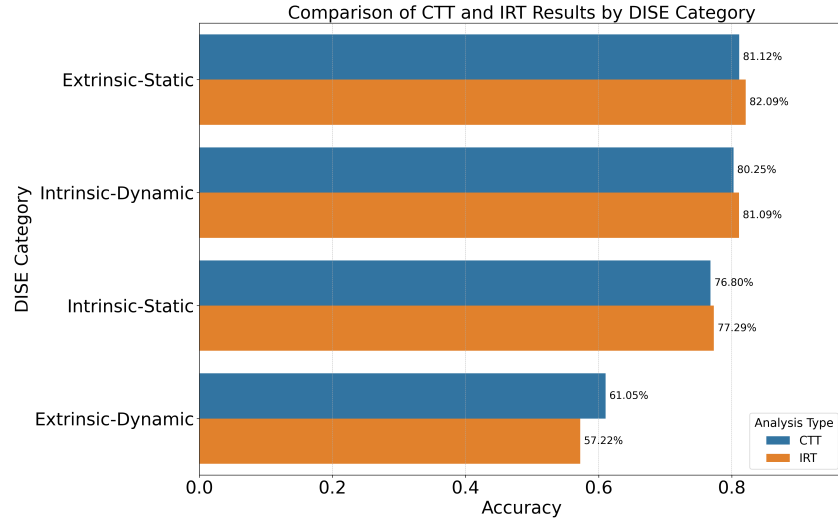
Table 11: Human Accuracy by DISE Category

DISE Category	CTT Accuracy (95% CI)	IRT Accuracy (95% CI)
Extrinsic-Dynamic	61.05% $\pm$ 4.46%	57.22% $\pm$ 8.67%
Extrinsic-Static	81.12% $\pm$ 4.38%	82.09% $\pm$ 7.02%
Intrinsic-Dynamic	80.25% $\pm$ 2.05%	81.09% $\pm$ 3.42%
Intrinsic-Static	76.80% $\pm$ 3.90%	77.29% $\pm$ 7.49%
Overall	76.84% $\pm$ 2.02%	76.92% $\pm$ 3.79%



(a) Human Accuracy by CTT.

(b) Human Accuracy in different tasks by CTT.



(c) Comparison of CTT and IRT results.

Figure 10: Human Performance Results by CTT and comparison of CTT and IRT results.